SOFTWARE ENGINEERING LAB 1

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Use-Case Flow Specification

Problem Statement #36 — Restaurant Table Booking & Pre-Ordering App

Lab 1: Requirements Engineering & UML Use-Case Modelling

User Case:	UC-02 reserve table &pre-order food items
Primary Actor	Diner
Secondary actors	Payment gateway (external systems), kitchen display system (external system)

Preconditions

1. Diner has a valid app account and is logged in.
2. Restaurant is open for reservations at the requested date/time, and the live floor-plan data is current.

Postconditions (Success)

1. The selected table is marked Reserved and linked to the diner's account for the chosen time slot.
2. A pre-order ticket, including the estimated kitchen preparation timeline, is queued in the Kitchen Display System.
3. Payment/deposit is captured and a confirmation with reservation details is sent to the diner.

Main Success Scenario

1. Diner opens the app and navigates to the live floor plan for the desired date and time.
2. System displays the interactive 2D floor plan showing which tables are open, reserved, or occupied.
3. Diner selects an open table.
4. System places a temporary hold on the table and prompts the diner to add a pre-order.
5. Diner browses the digital menu and selects food items to pre-order.
6. System calculates the estimated kitchen preparation timeline for the pre-order and displays it to the diner.
7. Diner reviews and confirms the reservation and pre-order.
8. System processes the payment/deposit via the Payment Gateway.
9. System confirms the reservation, queues the pre-order ticket to the Kitchen Display System, and syncs the table's status to Reserve across all manager terminals.
10. System sends the diner a confirmation with reservation details and the estimated prep/arrival timeline.
11. Use case ends successfully.

Alternate Flow — 8a: Payment Declined

- 8a1. System displays a payment-failed message to the diner.
- 8a2. Diner is prompted to retry with a different payment method.
- 8a3. If payment fails again after 2 attempts, the table hold is released, the pre-order is discarded, and the diner is notified that the reservation could not be completed.